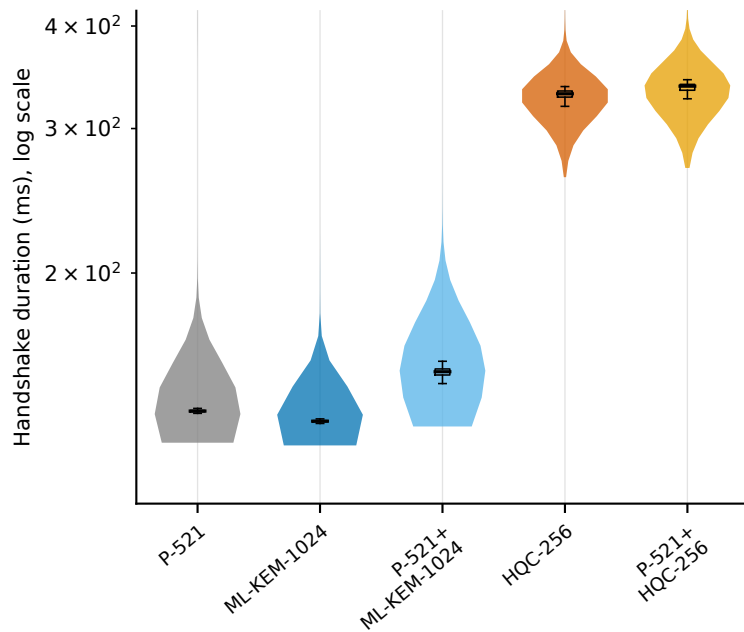
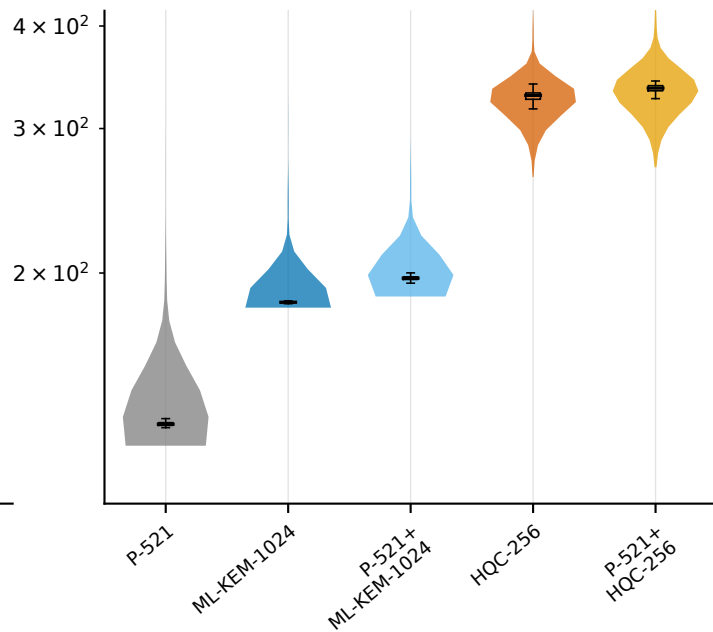


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	140.3	46.5	1.16k	136.8	48.4	1.15k
ML-KEM-1024	134.1	36.3	1.22k	188.3	26.1	1.20k
P-521+ML-KEM-1024	158.4	52.6	1.22k	200.2	25.8	1.34k
HQC-256	332.2	21.0	1.47k	328.6	16.1	1.48k
P-521+HQC-256	342.4	24.1	1.42k	338.6	19.2	1.33k